Tutorial for geochemical and reactive transport modelling using the HPx framework – Introducing biogeochemical reaction modelling using PHREEQC

3. DESCRIPTION OF THE CEMENT MODULE

3.1 Download and install

The cement module is available on github via this link. One need to download a zip file, unzip it and bring the module .

A module should be stored in the general Module directory of HPx (HYDRUS 5.x) and can be used in several projects. Tutorial 3-1 shows how to obtain the module and place it in the correct folder such that it can be activated in your projects.

TUTORIAL 3-1 INSTALL CEMENT MODULE

Description

To add the cement module to HPGeochemistry

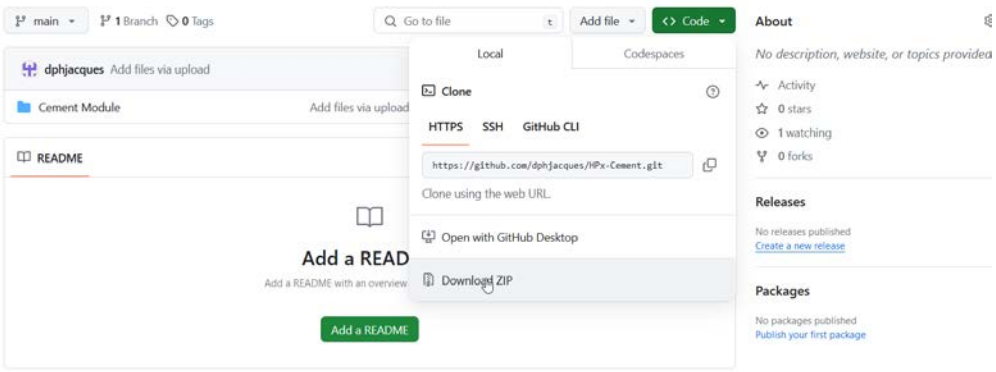
You will learn to

Add a module to HPGeochemistry

Download Cement Module

Go to the github repository via **link**

Download the repository via Code and Download ZIP



Find the general folder for Modules

The folders and files under Cement Module should be placed in the default Module folder of HPx/HYDRUS 5.x. It will then be available to be used it in different projects.

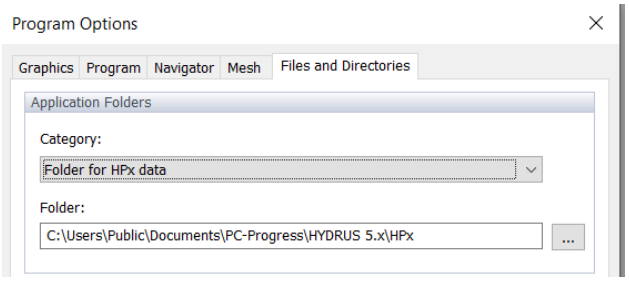
The default folder can be found via

(i) HYDRUS 5.0 GUI

Go to `Options->Program Options`...

Go to `Files and Directories` tab

Change the `Category` of the `Application Folders` to `Folder for HPx` data



The default folder is a subfolder of this folder. In this case, it is:

`C:\Users\Public\Documents\PC-Progress\HYDRUS 5.x\HPx\Modules`

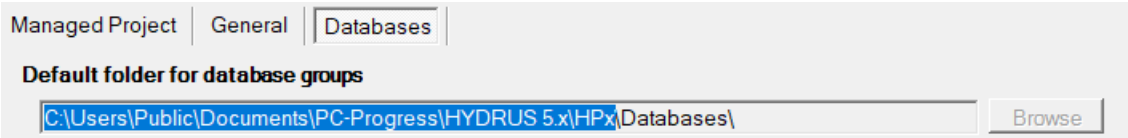
(ii) HPGeochemistry GUI

Go to `Settings and Preferences`

Go to the `Settings` tab

Go to the `Databases` subtab

The folder in the textbox "`Default folder for database groups`" indicates the default folder for databases. The parent folder gives also the path to subfolder `Modules`. Thus, in case you see



the default folder for modules is

`C:\Users\Public\Documents\PC-Progress\HYDRUS 5.x\HPx\Modules`

Place the cement module in the general folder

Make a subfolder `Cement Module` in the default folder for modules

Unzip the downloaded

Place the subfolder in the folder `HPx-Cement-main\Cement Module` in the subfolder `Cement Module`

END TUTORIAL 3-1

3.2 Templates

4.1 Set-up HPGeochemistry

The tutorial for setting up a project, add databases and modules script is limited to a list of instructions. The reader is referred to XX for more details on the basic use of HPGeochemistry.

All building blocks are present in the downloaded database files and cement module.

TUTORIAL 4-1 CREATE MANAGED PROJECT WITH THE CEMENT MODULE

Description

Create the project for this tutorial and add the cement module in the project

You will learn to

To create a managed project

To add a module to the project

Depends on

Tutorial 3-1 Install CEMENT module

Step 1 – Create project

Click `Project Manager` button

Click `Create new project` button

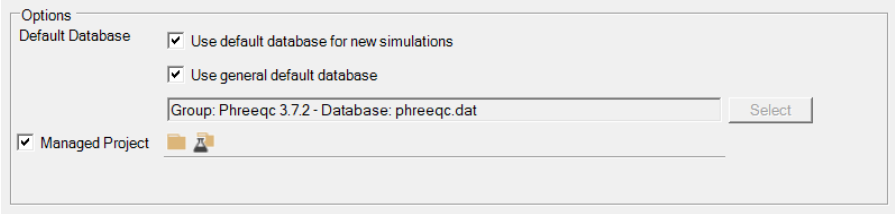
The `Metadata` dialogue window appears.

Give a name to your project

Step 2 – Add cement module to the project

A prerequisite to use a module in HPGeochemistry is that it is a managed project:

Ensure that the Managed Project is checked



For the Cement Module, folders for data, database and modules are required.

Open the `Folder Manager` for a managed project

Ensure that the folder `Data` (`04_data`) is checked

Options

Folders

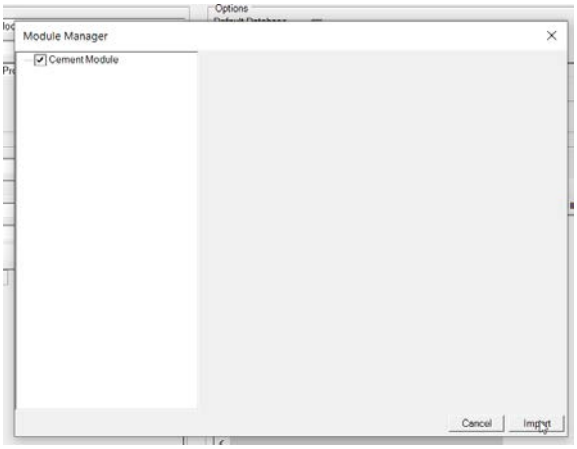
Name	Folder	Create	Include	Files
Background Documents	01_background-documents	☑	☐	☑
Literature Documents	02_literature	☑	☐	☑
Model	03_model-description	☑	☐	☑
Data	04_data	☑	☐	☑
Database	10_database	☑	☑	☐
Scripts	11_scripts	☑	☑	☐
Modules	12_modules	☑	☑	☐
GeoChemModels	20_geochemical-models	☑	☐	☐
Transport	21_transport-models	☑	☐	☐
Figures	50_figures	☐	☐	☑
Report	51_report	☐	☐	☑
Deprecated	98_deprecated-files	☐	☐	☑
Temporary	99_temporary-files	☐	☐	☑

OK

Open the `Module Manager` for a managed project 

Check `Cement Module`

Click `Import`



Click **OK**

END TUTORIAL 4-1

4.2 Modelling CEM-I hydrate assemblage and pore water composition

4.2.1 Calculation of the hydrate assemblage for a CEM-I oxide composition

TUTORIAL 4-2 CALCULATE THE CEMENT HYDRATES ASSEMBLAGE AND PORE WATER COMPOSITION FOR A CEM I

Description

Calculate the hydrates assemblage and pore water composition of the hydration of 100 g CEM-I with a water/cement ratio of 0.5 and at a temperature of 25°C. The CEM I oxide composition is taken from the Lothenbach and Winnefeld (2006) (their table 1). Calculations are done in the Ca-Si-Al/Fe-S-Mg-C-Na-K system.

You will learn to

Start a simulation from a template in a module

Depends on

Tutorial 4-1 Create managed project with the cement module

Step 1 – Create simulation from template for the calculation of the hydrates assemblages

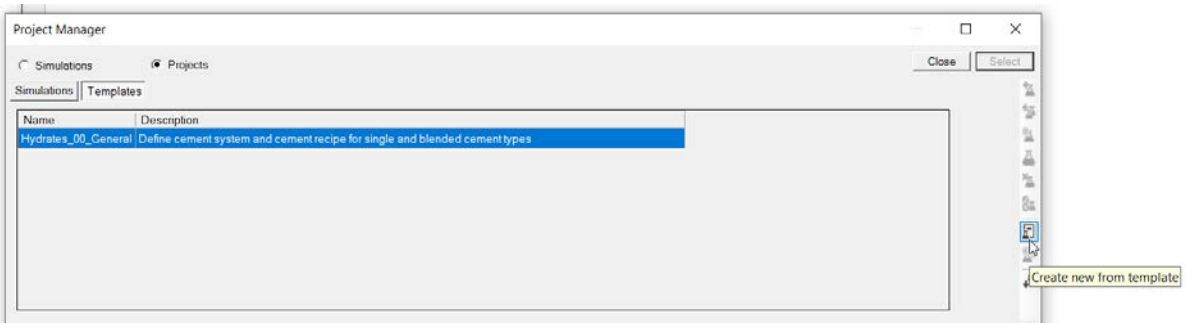
Go to `Project Manager`

Select the project in the bottom panel

Go to `Templates` tab in the top panel

Select the template "`Hydrates_00_general`"

Click the "`Create new from template`" button



The `Metadata` dialogue window opens.

Define the name of the simulation (`CEM I`)

Click `OK`

The Project Manager is active again.

Select the simulation (CEM I) and click `Select`

Step 2 – Define the problem

Only information in the input tables needs to be changed if values are different from the default values. All instructions and data are included in the simulation via the template.

Go to the `Global Definitions` input tab

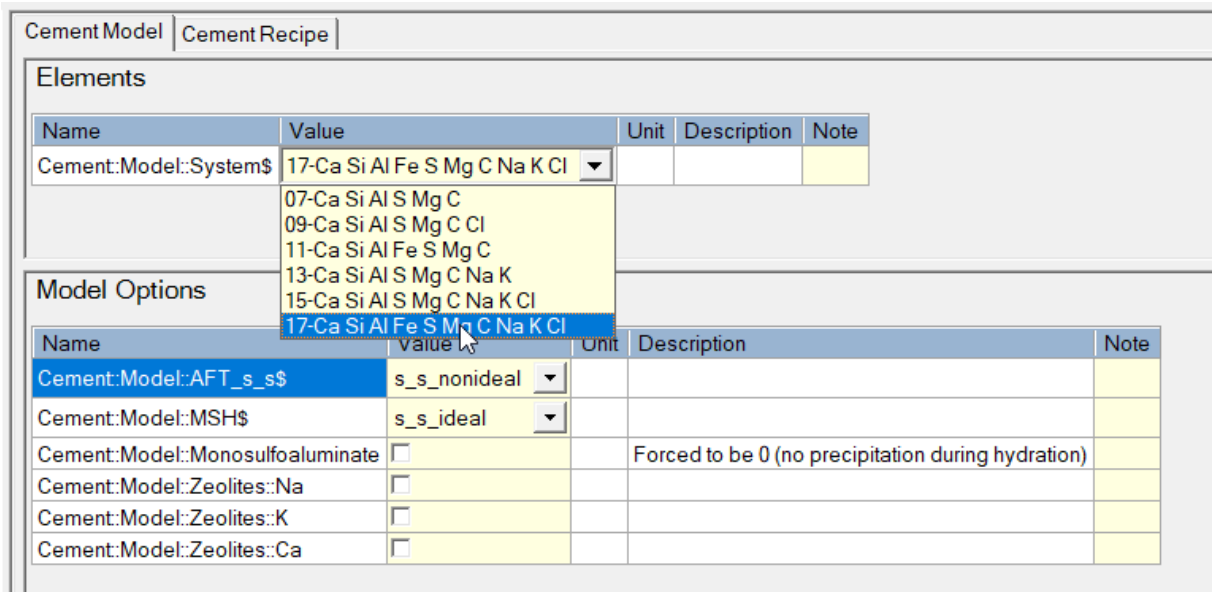
Go to the `Global Variables`

> The template consists of two input tables: "`Cement Model`" and "`Cement Recipe`" to, respectively, select the cement model and define the cement recipe.
>
> Go to the `Cement Model` tab

The first group "Elements" defines which elements are to be considered in the calculations. Selection is done via the drop down menu for the variable `Cement::Model::System$`.

> Select system 17 from the drop down menu. Note that this system also contains Cl but is neglected in this tutorial (no Cl in the clinker composition)

The second group "`Model Options`" allows to select some specific minerals to be included or excluded from the models – default selections are used in this tutorial.



> Go to the `Cement Recipe` tab

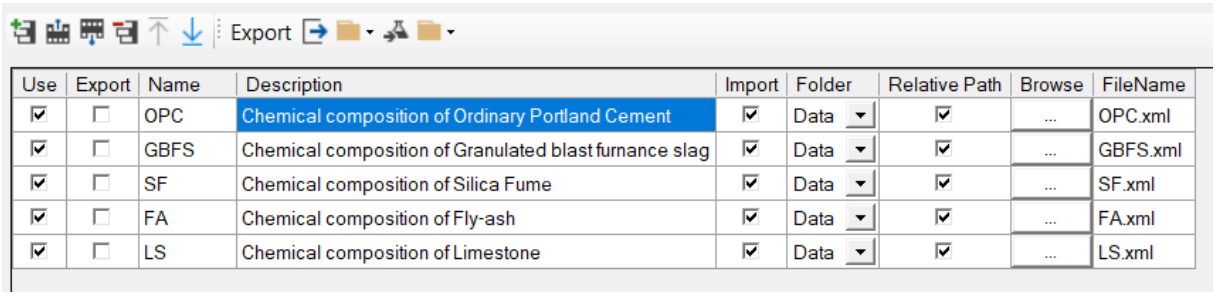
In the first group "Properties", the temperature (°C), water/cement ratio (-) and amount of cement (g) are given. In this tutorial, the default values are used.

The five other groups represent the ordinary Portland cement (OPC) clinker and following supplementary cementitious materials (SCMs): granulated blast furnace slag (GBFS), silica fume (SF), fly-ash (FA) and limestone (LS). For each material, it is possible to define oxide composition, the amount (mass%), the reacted amount (% reacted) and the bulk density (kg/dm^3).

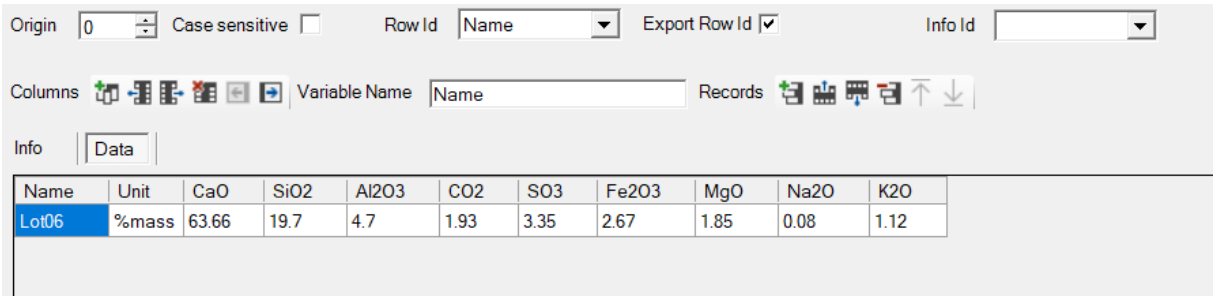
> For the cement recipe in this tutorial (CEM-I),

The composition of the OPC clinker and the SCMs are given via data tables defined in the DataBlock. Each record in the data table forms an item in the drop down list. Beside the default options, additional compositions can be defined (see Tutorial 4-3).

> Go to `DataBlock`



Select **OPC** data block in the top panel. Information in that data block is shown in the bottom panel:



For each of the cement constituents (OPC and SCMs), a data table is available. In each of these data table, an oxide composition is given by default (see example for OPC). The indicator in the column `Name` is used in the drop down list. The reference to the oxide composition is listed in the `Info` tab. Other compositions can be added (see Tutorial 4-3).

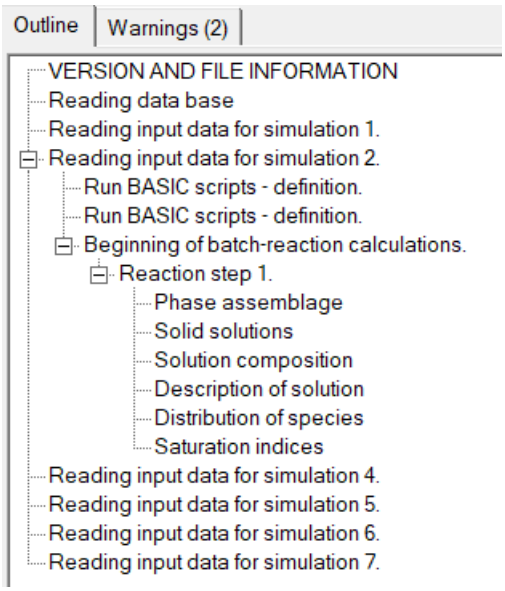
Run the problem 

Step 3 – Check the output

Go to `phreeqc.out` file

Go to `Outline` tab

Go to `Reading input data for simulation 2` -> `Beginning of batch-reaction calculation`



Note first that there are warnings. This is because Cl-containing hydrates are in the hydrate assemblage (due to the selection of system 17), but there is no Cl in the input.

The output file gives the standard phreeqc output related to a batch reaction calculation. All geochemical information can be obtained from this file. The template also produces a special output file with the most important information for the cement system.

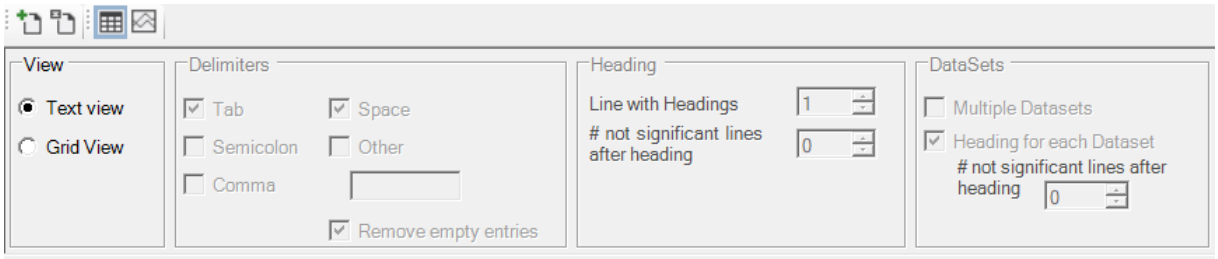
Go to the Data File output workspace

Click Open Data File

Open the file "Cement_Hydration_1001.out"

Go to the View options

Select Text View



Cement_Hydration_1001.out

```
CEMENT RECIPE
                        OPC clinkers            100 weight%
                         Reacted OPC            100 % reacted

                    Amount of cement            100
                           w/c ratio           0.50
                         Temperature             25

CONTENT
             Name          Moles
              CaO       1.14e+00
              SiO2      3.28e-01
             Al2O3      4.61e-02
              SO3       4.18e-02
              CO2       4.39e-02
              MgO       4.59e-02
             Fe2O3      1.67e-02
              Na2O      1.29e-03
              K2O       1.19e-02

HYDRATED PHASES

             Name          Moles          Grams          Volume

      Portlandite       4.17e-01          30.90           13.79
              Cal       3.49e-02           3.50            1.29
    hydrotalcite       1.15e-02           5.09            2.53
   monocarbonate       5.04e-03           2.87            1.32
              AFT       4.56e-02          18.95           10.68
              CSH       4.28e-01          61.52           27.22
   HydrogarnetFe       3.11e-02          13.35            4.53
              MSH       9.51e-16           0.00            0.00
```

In this view, you find information on the cement recipe, the oxide composition, the hydrated phases and the pore water composition, calculated volumes, and information on solid solution composition.

END TUTORIAL 4-2